

Is cosmic fate an inference or an assumption?

A locally preregistered fragility audit of cosmic-fate inference with DESI DR2, Pantheon+, and *Planck*

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Abstract. The DESI era has renewed interest in the ultimate fate of the Universe. We ask how much of a fate verdict is constrained by data and how much is induced by assumptions. We combine DESI DR2 BAO, *Planck* 2018 distance priors, and Pantheon+/SH0ES supernovae under a locally preregistered analysis plan, propagate posterior samples through a frozen fate classifier, calibrate the verdict with repeated data, and vary its prior, statistic, parametrization, and future-continuation rule. Within CPL and the baseline prior, the posterior Big Rip probability is 0.18%, with a nested-sampling point estimate of 0.179%. Across 500 fitted-truth Λ CDM mocks, the heat-death-compatible posterior mass is consistent with a uniform distribution; 2 mocks are at least as deep in the anti-rip tail as the data, giving a plus-one estimate of 0.00599 with an exact 95% interval of 0.000485–0.0144. A quintessence prior excludes the rip before data are considered. Four converged native-prior parametrizations yield Big Rip probabilities from 0.21% to 49.96%, while their induced prior rip masses already span 25.8%–58.4%. This is a sensitivity audit, not a common-prior model comparison: the four pushforwards occupy incompatible supports in function space. A calibrated seven-node reconstruction leaves its final future node essentially prior-dominated. Holding the entire history up to the present, all current-background observables, and the likelihood fixed, four registered continuation families give a descriptive Big Rip envelope of 0.00%–50.63%; 100% of source-history weight admits both rip and heat-death-compatible continuations. These kinematic continuations are not asserted to be stable covariant micro-physical theories, and no probability is assigned to them. Current observations can therefore yield sharp fate probabilities within an adopted model, but do not identify a unique infinite future without a physical continuation law. All quantitative results are background-level and conditional on the compressed-CMB likelihood.

Keywords: dark energy theory; Bayesian reasoning; statistical sampling techniques; baryon acoustic oscillations

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1 Introduction

The DESI era has revived the oldest question in cosmology. Baryon acoustic oscillation measurements from DESI DR2, combined with cosmic microwave background (CMB) and Type Ia supernova data, favor solutions in the “thawing” quadrant $w_0 > -1$, $w_a < 0$ [1]. The reported strength is analysis-dependent. The original official combinations gave 2.8–4.2 σ ;

subsequent supernova recalibration brings corrected Pantheon+, Union3.1, and DES-Dovekie combinations into the narrower $3.2\text{--}3.4\sigma$ range, while fully Bayesian evidence calculations can modestly favor Λ CDM after the extended model’s prior volume is included [1–8]. After this analysis freeze, DESI added a one-percent Alcock–Paczyński measurement from the same DR2 $\text{Ly}\alpha$ forest used for its BAO point. The joint $\text{Ly}\alpha$ AP+BAO update reports 2.7σ for DESI+CMB and 3.2σ after adding supernovae, slightly reducing rather than eliminating the dynamical-dark-energy discrepancy [9]. Specific parametrizations and physical models have long been translated into future-fate statements, including work designed explicitly to avoid CPL’s unbounded future behavior [10–12].

This paper asks a different question. Rather than “what is the fate?”, we ask: *of the fate verdict, how much is forced by the data, and how much is donated by the assumptions?* The question has a distinguished qualitative pedigree—Krauss and Turner [13] argued a quarter-century ago that dark energy severs the link between geometry and destiny, and Kallosh and Linde [14] computed observational doomsday bounds for specific potentials. We are unaware of a published study that combines an end-to-end repeated- Λ CDM-mock calibration of a posterior fate-class probability with a finite sensitivity audit over the eight analysis axes declared here. This is the paper’s scoped methodological contribution, not a claim that earlier work ignored model dependence.

The paper separates three questions. First, within a specified CPL analysis, what fate probability is obtained and how does that statistic behave under repeated Λ CDM data? Second, how does the verdict respond to data choices, priors, inference statistics, and four commonly used parametrizations? Third, after the entire observed history is fixed, what remains unidentified about its continuation? This ordering separates a model-conditional numerical result from the broader question of whether present data license an infinite- future extrapolation.

Three results organize the answer. Within continuous CPL, the direction-class posterior mass is consistent with $U(0, 1)$ when the truth is exactly on the Λ CDM fate boundary, while the real data lie unusually deep in its anti-rip tail. Across native-prior parametrizations, similar descriptive in-window fit quality coexists with posterior Big Rip probabilities from approximately 0.2% to 50%; these are complete-analysis sensitivity diagnostics rather than a common-prior model competition. Finally, a calibrated seven-node fit and an exact fixed-history continuation audit show that apparent future-node information can be transported by prior correlations and that the registered continuations of one observed history support both thermodynamic sides. The last statement is a partial-identification result within a declared kinematic class, not an impossibility theorem for all microphysical theories.

We therefore report current fate estimates as model-conditional predictions whose structural uncertainty is not contained in their within-model posterior intervals. The contribution is a locally preregistered, repeated-data- calibrated sensitivity protocol that makes the conditioning explicit.

2 Preregistered inference framework

The analysis plan specified the model space, priors, dataset combinations, criteria for classifying fates, statistical rules, and pass/fail gates. The protocol was written and frozen in local version control before the first target fate calculation, according to the preserved Git history. Because the repository was made public only afterwards, the original freeze lacks an independently verifiable timestamp. We therefore describe the study as locally prereg-

istered, not as a registry-verified preregistration. Specifically, commit 98abc20 under tag `prereg-v1`, recorded at 10:26 local time on 2026-07-03, is an ancestor of the first real-data fate-result commit recorded at 11:29. The frozen plan is reproduced in English translation in Appendix A; commit ancestry, timestamps, tree identifiers, and the plan checksum are listed in the archived provenance audit, `plan/PREREGISTRATION_PROVENANCE.md`.

More precisely, this is a secondary-data preregistration of the fate-translation audit. The datasets and validation strategy were already known, but the core classifier, audit axes, gates, and two reportable outcome forms were frozen before the first target fate calculation. It is not a claim of preregistration before data access or of an independently timestamped public registration.

The 500-mock, off-boundary-power, seven-node, and fixed-history continuation results were not part of the original local preregistration. They belong to a separately versioned strengthening protocol, frozen before the corresponding results. The immutable historical checkpoint is tag `paper-v1x-final-20260722`; extension products are isolated below `runs/prd_extension/`. We describe this layer as a prospectively specified public extension, not as part of the original preregistered core. Its append-only ledger records execution-order deviations and corrective scope changes.

Departures from the frozen plan are recorded in an append-only log (Appendix B) stating what changed, why, and when relative to data contact. Three amendments to the frozen core occurred: A-001 recalibrated the pipeline-validation benchmark after we established that a compressed-CMB likelihood cannot reproduce the full-likelihood exclusion significance (Sec. 2.1); A-002 replaced a mock-recovery criterion that is mathematically unsatisfiable for a truth on a fate boundary (Sec. 4); and A-003 added an analytic fate branch after the frozen OTHER-budget audit was triggered by bounded-future parametrizations (Sec. 5.4). In each case the trigger, argument, and sign-off are logged. Four later records are explicitly post-hoc: A-004 tests whether the grammar-specific priors can be matched by importance reweighting; A-005 corrects the finite-limit fate semantics without altering CPL-based results; A-006 adds an early-dark-energy hard gate; and A-007 documents the model-specific translation of P1’s early-matter-domination intent without retroactively treating that implementation detail as explicitly preregistered.

The data also predate the plan: DESI DR2 constraints were public before the freeze. The version-controlled record documents analysis-choice discipline—a declared classifier, gates, fragility metrics, and two equally reportable outcome forms—rather than prediction novelty. The published w_0 – w_a constraint is used only as a Gate-1 validation target. Six analytical checks for the null campaign were recorded after approximately 17 of 100 mocks and before the remaining campaign completed; they are prospective checks for the remaining runs, not pre-analysis registrations. The zero-residual prediction was recorded before that run (Appendix E). This discipline is weaker than survey blinding, which can shield catalogs or data vectors from the analyst [15, 16]; it makes the workflow auditable but does not eliminate all analyst degrees of freedom. The methodological contribution is the resulting transparent sensitivity audit, not an external-registry claim.

2.1 Data, likelihoods, and validation

We use three publicly released datasets: DESI DR2 BAO compressed measurements (13 points over seven redshift bins, $0.295 \leq z \leq 2.33$) [1]; the Pantheon+ Type Ia supernova compilation with full stat+sys covariance, with and without the SH0ES Cepheid anchor [17–19]; and *Planck* 2018 information [20] compressed into the distance priors $(R, \ell_A, \Omega_b h^2)$ of

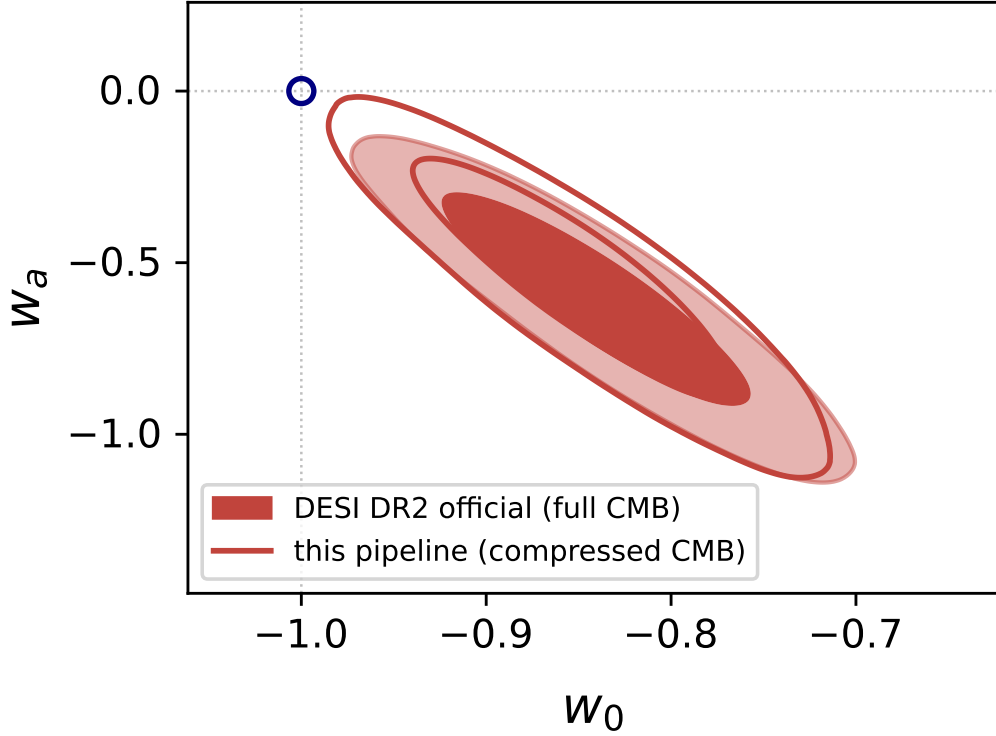


Figure 1. Gate-1 chain-level validation: this pipeline (compressed CMB, open contours) versus the official DESI DR2 chains (full CMB + lensing, filled). All marginal shifts $\leq 0.33\sigma$. Here and throughout, contour pairs show the 68% and 95% credible regions.

Ref. [21], validated there for CPL-type models. Files, versions, and checksums are registered in the repository manifest (Appendix G).

Two validation results define what the pipeline can claim. First, against the official DESI DR2 w_0w_a CDM chains, our marginals agree to $\leq 0.33\sigma$ in every parameter, with Ω_m agreeing to $< 0.1\sigma$ and H_0 to $< 0.25\sigma$ (Fig. 1). Second, the compressed CMB prior carries a known, stable information deficit: our Λ CDM exclusion is 2.25σ ($\Delta\chi^2 = 7.44$) where the full-likelihood value is 2.8σ , and the same $\approx 0.5\sigma$ discount recurs across all three supernova compilations (Sec. 5.1), consistent with independent compressed-likelihood analyses [5]. Every significance below inherits the compressed-CMB condition; we price it rather than hide it. One procedural lesson is worth recording: compressed priors must be predicted with the same z_*/r_s conventions used at extraction—mixing conventions shifted ℓ_A by 3σ of pure artifact in an early pipeline version, caught by a known-answer test before any science ran.

2.2 Fate taxonomy and classifier

Each posterior sample defines a background

$$\frac{H^2(a)}{H_0^2} = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{\text{DE}} f_{\text{DE}}(a), \quad (2.1)$$

integrated from $a = 1$ to $a_{\text{max}} = 10^4$ and classified with frozen criteria, in priority order: CRUNCH ($H^2 \leq 0$ reached); RIP (finite-time singularity [22]: convergent $\int da/aH$; for

Table 1. Fate probabilities, D0+CPL+P1. *Caution: the direction component of this verdict is uninformative under the null (Sec. 4).*

Class	P	thermodynamic class
DECAY	$99.82\% \pm 0.06\%$	heat-death-compatible
RIP	$0.18\% \pm 0.06\%$ (nested 0.179%)	non-heat-death
DS	0 (grammar artifact; Sec. 5.4)	heat-death-compatible
OTHER/boundary	0 / 0	—

parametrizations whose $w(a)$ has a finite limit w_∞ , A-003 introduced an analytic branch and A-005 corrects its physical boundary to $w_\infty < -1$); DS (exact de Sitter asymptotics, $w_\infty = -1$ with persistent dark energy, in the finite-limit branch); DECAY (dark-energy dilution); OTHER (unclassifiable—audited under a frozen 1% budget whose breach triggers the amendment process, as occurred for A-003). Every class carries a boundary flag: numerical-branch samples that flip when thresholds are doubled or halved, and finite-limit samples satisfying $|w_\infty + 1| \leq 0.01$, are reported as boundary-adjacent without changing their physical label.

For this audit, DS and DECAY are grouped into a coarse heat-death-compatible macro-class because both avoid a finite-time singularity and expand forever in the flat backgrounds considered here. This operational grouping does not assert thermodynamic equivalence: horizon structure, temperature floors, and accessible free energy can differ between the two. Within flat CPL backgrounds the only non-heat-death branch is the Big Rip. The classifier passes a ten-case unit suite spanning all classes, boundary behavior, and two documented numerical edge cases (Appendix D).

3 Conditional fate inference in CPL

Figure 2 shows the D0 (DESI DR2 + CMB prior + Pantheon+SH0ES) CPL [23, 24] posterior, with

$$\begin{aligned} w_0 &= -0.884 \pm 0.057, & w_a &= -0.62^{+0.25}_{-0.22}, \\ \Omega_m &= 0.2997 \pm 0.0050, & H_0 &= 69.00 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}, \end{aligned}$$

from four chains with $R-1 = 0.0033$, drawn over the fate partition of the (w_0, w_a) plane. The posterior presses against the zero-width RIP/DECAY boundary at $w_a = 0$; the consequences of that geometry occupy the next section.

Propagating every sample through the classifier yields Table 1: $P(\text{DECAY}) = 99.82\% \pm 0.06\%$, $P(\text{RIP}) = 0.18\% \pm 0.06\%$ (batch-means errors), zero OTHER, zero boundary flags. Under the frozen rule that sub-percent tails require a second sampling calculation, nested sampling gives a consistent weighted point estimate, $P(\text{RIP}) = 0.179\%$. Across three independent seeds, the between-seed standard deviation is 0.039 percentage points and the range is 0.135–0.209%. The previously quoted 0.038% binomial standard error describes only a fixed-seed equal-weight resample; it is not run-to-run nested-sampling uncertainty and is therefore not used as an error bar. The weighted nested OTHER mass is 0.0093%, below the registered budget.

Two seeds of unease belong here. First, for CPL against ΛCDM , the three-seed mean is $\ln B = -1.617$. The between-seed SD is 0.224, with range $[-1.865, -1.431]$. The three per-run internal dynesty errors span 0.313–0.314. Bayesian evidence *favours* ΛCDM while

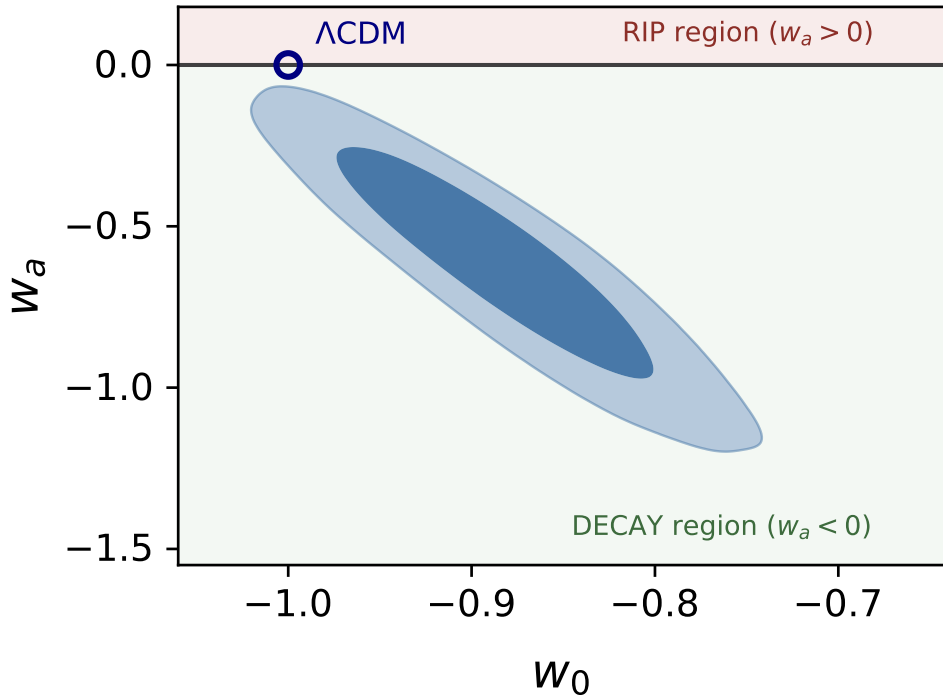


Figure 2. D0+CPL+P1 posterior over the fate partition. The fate boundary $w_a = 0$ has zero width; Λ CDM sits exactly on it.

$\Delta\chi^2$ disfavors it at 2.25σ —a live Lindley phenomenon [25], developed as a fragility axis in Sec. 5.3. Second, $P(\text{DS}) = 0$ exactly: strict de Sitter asymptotics is a measure-zero locus of CPL parameter space, so this zero was decided by the parametrization before any photon arrived. Before interpreting Table 1 at all, however, one must ask what such a table looks like when the truth is exactly Λ CDM—the subject of the next section.

4 Repeated-data calibration

4.1 Boundary-truth calibration

We are unaware of a published repeated- Λ CDM-mock calibration of a posterior cosmic-fate class probability using the same end-to-end inference pipeline. We first generated 100 mock datasets whose truth is the Λ CDM best fit to D0. A separately frozen extension appended 400 without optional stopping, giving 500 noisy realizations in total. Noise is drawn from the exact likelihood covariances (SN 1657×1657 ; BAO 13×13 ; CMB prior 3×3), and every mock is processed by the same CPL+P1 pipeline. A separate zero-residual realization, excluded from the finite null distribution, makes all three χ^2 values vanish at the truth and closes the mock-likelihood loop.

Because the Λ CDM truth sits on the zero-width CPL fate boundary, local regular asymptotics predict the heat-death-compatible posterior mass to approach $U(0,1)$ across repeated realizations: the local likelihood must be calibrated, the boundary locally linear, and the prior absolutely continuous with no point mass on Λ CDM. Under those conditions a direction verdict is correct in $\sim 50\%$ of mocks. This is a property of the specified deci-

Table 2. Registered truth-specific power. $\Delta\chi^2$ is relative to the best of the six constrained D0 truth fits, not a common distance from the fate boundary. Brackets are exact 95% binomial intervals. The “false” column is the fraction placing majority mass on the opposite side.

Truth	w_a	$\Delta\chi^2$	correct side	direction power	depth power	false-sign fraction
wam060	−0.60	0.00	heat	1.00 [0.964,1.000]	0.90 [0.824,0.951]	0.00 [0.000,0.036]
wam030	−0.30	1.67	heat	0.93 [0.861,0.971]	0.47 [0.369,0.572]	0.07 [0.029,0.139]
wam015	−0.15	4.05	heat	0.84 [0.753,0.906]	0.33 [0.239,0.431]	0.16 [0.094,0.247]
wap015	+0.15	12.71	RIP	0.77 [0.675,0.848]	0.18 [0.110,0.269]	0.23 [0.152,0.325]
wap030	+0.30	19.45	RIP	0.93 [0.861,0.971]	0.57 [0.467,0.669]	0.07 [0.029,0.139]
wap060	+0.60	39.37	RIP	1.00 [0.964,1.000]	0.98 [0.930,0.998]	0.00 [0.000,0.036]

sion problem, not a statement that pipeline quality is irrelevant and not a universal limit on Bayesian procedures. Spike-and-slab priors, explicit Λ CDM model probability, curved boundaries, or misspecified posterior widths change the result (Appendix F). These predictions were logged mid-campaign (at ~ 17 of 100 mocks), before the original campaign completed and before the zero-residual run. They all passed at the historical 100-mock checkpoint. The public extension did not assume that its original 0/100 lower-tail count would remain zero. In the pooled campaign, 264/500 (52.8%, exact 95% interval [48.3, 57.2]%) have $P(\text{heat}) > 0.5$, and the complete $P(\text{heat})$ distribution remains compatible with $U(0, 1)$ (KS $p = 0.154$; Fig. 3). The maximum OTHER and boundary fractions are 0.254% and 0.139%, respectively. The zero-residual mock returns 52/48. Its data vector equals the truth, but its likelihood covariance remains finite; it is a closed-loop diagnostic, not a zero-uncertainty experiment. The result isolates the role of boundary geometry within this continuous CPL analysis.

The null distribution also supplies the calibrated reading of the real data. The minimum null Big Rip probability is now 0.166%, slightly below the observed 0.18%; 2/500 mocks satisfy $P(\text{RIP})_{\text{mock}} \leq P(\text{RIP})_{\text{obs}}$. Thus the historical v1.x statement “below every realization” was correct for its first 100 mocks but does not survive the higher-resolution extension. The registered conditional plus-one value is $\hat{p} = (2 + 1)/(500 + 1) = 0.00599$, with the exact two-sided 95% Clopper–Pearson interval [0.000485, 0.0144] for the underlying fitted-truth tail probability. The asymptotic Wilks route gives $p = 0.024$, but the statistics are not equivalent: the mock campaign ranks the one-sided posterior mass $P(w_a > 0 \mid D)$ within a CPL pipeline, whereas Wilks tests the point $(-1, 0)$ against a two-parameter CPL alternative. Their numerical agreement is a useful qualitative cross-check, not two independent measurements of the same tail. The correct summary of Table 1 is therefore two-layered: *the CPL direction probability has the registered uniform null distribution, while the real data lie unusually deep in its anti-rip tail and also disfavor point- Λ CDM by a distinct likelihood-ratio test*. What that signal does and does not imply about the actual fate is the subject of the fragility matrix.

4.2 Off-boundary power

Null calibration at the $w_a = 0$ boundary does not show how quickly the statistic becomes informative away from that boundary. The extension therefore froze six CPL truths at $w_a = \{-0.60, -0.30, -0.15, +0.15, +0.30, +0.60\}$. At each value all other parameters were reprofiled against D0 before any of its 100 noisy mocks were generated. Direction power is the fraction placing more than one half of the posterior on the truth-consistent thermodynamic side. One-sided depth power is the fraction rejected at $\alpha = 0.05$ against the completed 500-mock null CDF. Table 2 reports exact binomial intervals.

Both registered outer-point alternatives pass the predeclared 0.80 direction gate, so the

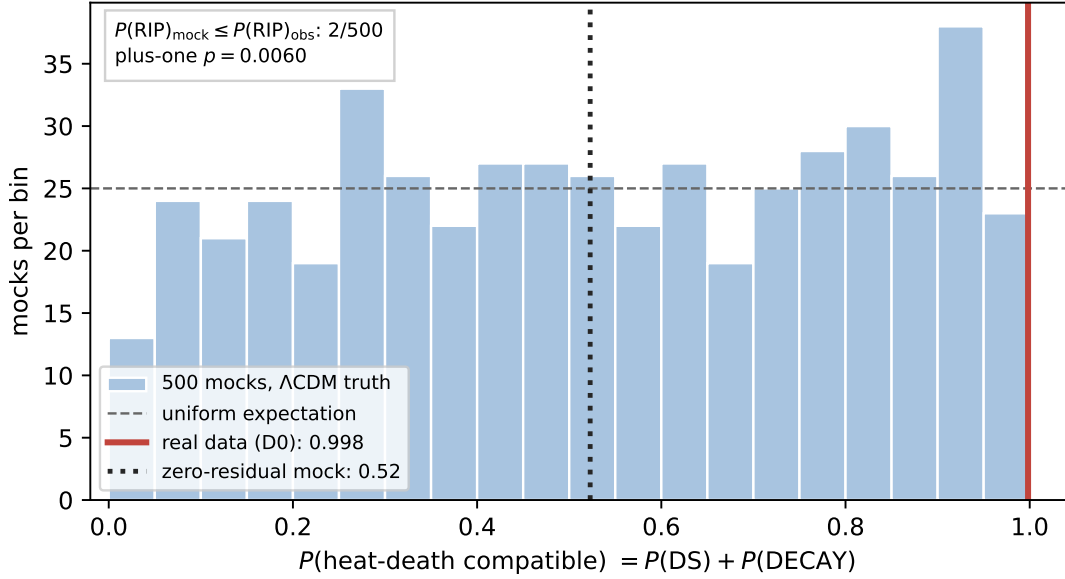


Figure 3. The completed 500-mock null calibration. Under Λ CDM truth the heat-death-compatible posterior mass is consistent with $U(0, 1)$ (KS $p = 0.154$) for continuous CPL+P1; the excluded zero-residual diagnostic returns 52/48. Two mocks have $P(\text{RIP}) \leq P(\text{RIP})_{\text{obs}}$; the red line marks the observed heat-death-compatible mass.

justified positive claim is that the direction classifier is powerful under those two alternatives. This is not a global statement: direction power at $w_a = +0.15$ is 0.77, and inner-point depth power is only 0.18–0.33. Moreover, the six generating truths have observed-D0 profile penalties from 0 to 39.37. Figure 4 therefore shows truth-specific operating characteristics, not a symmetric effect-size curve. The registered monotonicity rule merely checks for a reversal with disjoint exact intervals; no such reversal appears, but that coarse clear result is not separate evidence of power.

5 Sensitivity to analysis assumptions

5.1 Data axis

Table 3 and Fig. 5 show the leave-one-out response. The thawing direction ($w_0 > -1$, $w_a < 0$) survives every ablation—including removal of the CMB (D2; the CMB prior is replaced by a BBN baryon-density prior [26]), which *deepens* rather than erases the preference, a point against the hypothesis that the evidence is purely CMB–low- z tension. The magnitude, by contrast, is fragile: w_a swings threefold (-0.55 to -1.72), and the full combination is more moderate than any probe-removal subset (D2–D4)—the widely quoted $w_a \approx -0.6$ reflects partially offsetting dataset pulls. Swapping the supernova compilation (same model, same priors, same BAO and CMB) moves the Λ CDM exclusion from 2.25σ (Pantheon+) through 2.81σ (DES-Dovekie) to 3.27σ (Union3 [27]), reproducing the literature’s significance ladder [6, 7] inside a single pipeline; the SH0ES anchor alone moves H_0 by $+1.37 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and w_0 by -0.036 —Hubble-tension leakage into the dark-energy sector. Throughout, $P(\text{RIP})$ stays within 0.0008–0.376%: the direction-level conclusion remains sub-percent for every com-

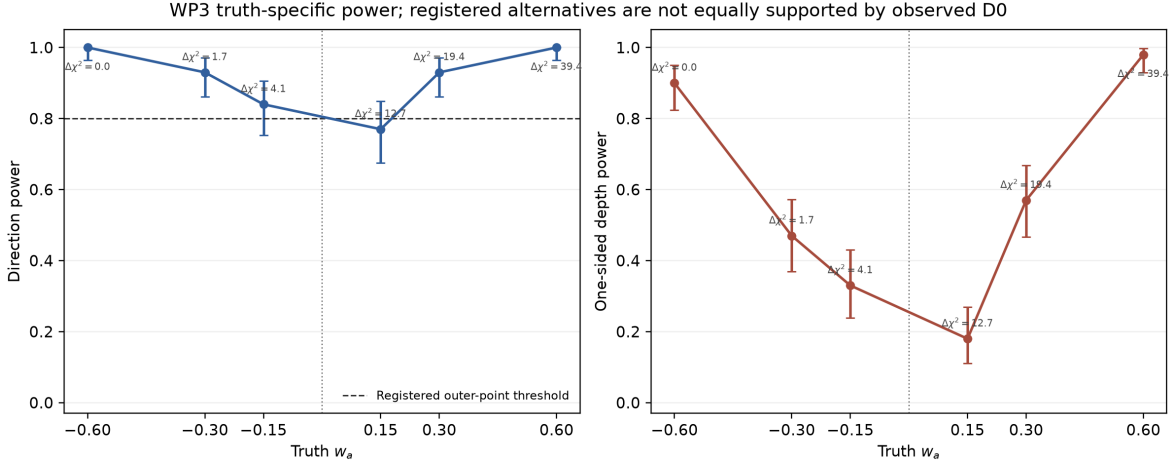


Figure 4. Direction and one-sided depth power at six registered constrained truths. Error bars are exact 95% binomial intervals; annotations give each truth’s D0 profile $\Delta\chi^2$ relative to the best registered truth. The unequal profile penalties prevent reading the two sides as equally supported, symmetric effect sizes.

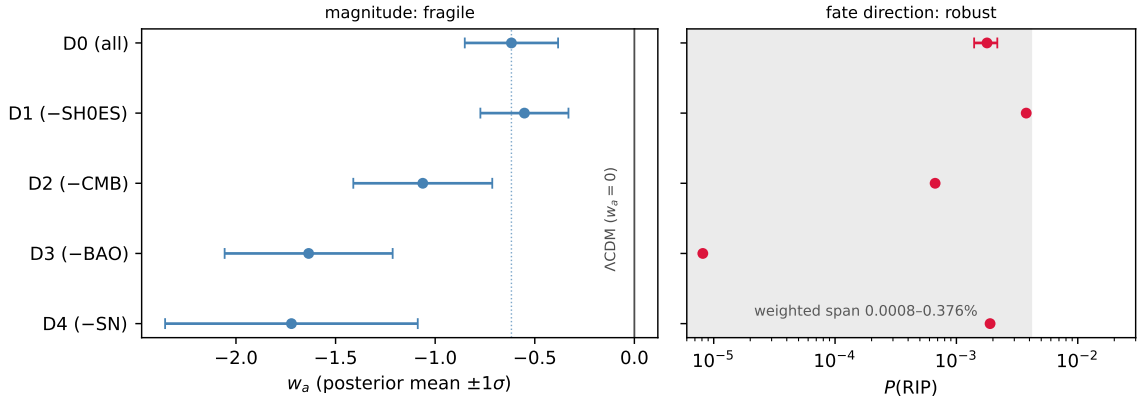


Figure 5. Leave-one-out forest (CPL+P1 fixed). Left: the w_a magnitude swings threefold across dataset ablations, and the full combination (dotted line) is more moderate than any probe-removal subset (D2–D4). Right: $P(\text{RIP})$ stays within 0.0008–0.376% throughout (shaded); the D3 entry is the small weighted tail retained by the original nested weights even though its fixed-seed equal-weight diagnostic contains zero RIP draws.

bination, but the numerical tail spans 2.665 dex and is therefore not robust in a multiplicative sense. Grammar dependence is larger still (Sec. 5.4).

5.2 Prior axis

Under the physical priors the question dissolves before data arrive: P2 (pointwise $w(a) \geq -1$, quintessence) and P3 (P2 + thawing) force $P(\text{RIP}) = 0$ *identically*—reported, per the frozen plan, as “excluded by prior.” The posterior is pinned into a sliver against the $w = -1$ wall ($w_0 = -0.975$, $w_a = -0.015$), and the wall costs the fit $\Delta\chi^2 = +8.9$ relative to P1: a clean measure of the current tension between the data and the quintessence hypothesis. P3 adds

Table 3. Leave-one-out (CPL+P1 fixed). Per the frozen sub-percent rule, the sub-percent entries were recomputed with nested sampling. The weighted point estimates are consistent in direction with the MCMC values; for D3 the original weights resolve a small positive tail that the equal-weight diagnostic rounds to zero. Binomial errors stored with the equal-weight resamples are numerical resampling diagnostics, not repeated nested-run uncertainties (per-combination records in the repository).

Combo	removed	w_0	w_a	$P(\text{RIP})$
D0	—	-0.884 ± 0.057	-0.62	0.18%
D1	SH0ES	-0.848 ± 0.055	-0.55	0.376%
D2	CMB	-0.853 ± 0.067	-1.06	0.067%
D3	BAO	-0.711 ± 0.083	-1.64	0.0008%
D4	SN	-0.43 ± 0.22	-1.72	0.189%

nothing to P2 on these data—its extra thawing assumptions are redundant once the pointwise bound binds. The prior axis reading is therefore maximal: $\log_{10} P(\text{RIP})$ spans from -2.7 to $-\infty$.

5.3 Inference-statistic axis

The same data, model, and priors yield the fit improvement $\Delta\chi^2 \equiv \chi^2_{\Lambda\text{CDM}} - \chi^2_{\text{CPL}} = 7.44$ (Wilks $p = 0.024$, disfavoring ΛCDM) and a three-seed $\ln B = -1.617$ (between-seed SD 0.224; Jeffreys-scale support *for* ΛCDM). Nothing is inconsistent: the two statistics ask different questions, and the Occam penalty of the 15.5-unit (w_0, w_a) prior area dominates the fit improvement. But the practical consequence is stark—*whether “the evidence favors dynamical dark energy” is true depends on which inference statistic one grants authority*, and the Bayes factor’s first-order dependence on prior volume makes it a fragility axis of its own, rarely reported as such. The registered information criteria split along the same fault line: $\Delta\text{AIC} = -\Delta\chi^2 + 2\Delta k = -3.4$ sides with the fit, while $\Delta\text{BIC} = -\Delta\chi^2 + \Delta k \ln N = +7.4$ sides with the evidence ($\Delta k = 2$, $N = 1673$ data points; negative values favor CPL)—the direction flip is reproduced within the information-criterion family itself.

An explicitly model-averaged reading makes the conditioning visible. As an *exploratory, post-hoc* diagnostic, assign equal prior odds to the two-model set $\{\Lambda\text{CDM}, \text{CPL}\}$. The measured Bayes factor then gives $P(\Lambda\text{CDM} \mid D) = 83.4\%$ and $P(\text{CPL} \mid D) = 16.6\%$; because exact ΛCDM is heat-death-compatible, the corresponding model-averaged compatibility is 99.97%. These values are not privileged—they change with the model set and model prior odds—but they demonstrate why the continuous-CPL 50/50 boundary result cannot be promoted to a universal Bayesian claim.

5.4 Parametrization axis

The four converged parametrizations are sensitivity probes, not contestants in a common-prior model comparison. Their archived chain-minimum AIC values are descriptively similar (Appendix C), but those minima are neither dedicated optimizations nor model evidences. Related in-window degeneracies are known for overlapping two-parameter families [28]; they do not establish equal model probability for this set.

The four fits do *not* have identical priors. Their coordinate dimensions, early-matter-domination constraints, and induced measures on functions and fates differ. A deterministic P1 prior audit makes the baseline explicit: prior $P(\text{RIP})$ is 25.8% (CPL), 40.0% (JBP), 58.4% (BA), and 50.6% (BIN4). The likelihood strongly suppresses the rip in CPL, JBP,

Table 4. Grammar-specific induced prior fate mass and posterior fate probabilities (D0 fixed; A-005 corrected semantics). The CPL posterior row uses the generic cross-grammar background machinery and agrees with the baseline within Monte Carlo error. The rows use different native coordinate priors and are a sensitivity audit, not a common-prior model comparison.

Grammar	prior $P(\text{RIP})$	posterior $P(\text{RIP})$	posterior $P(\text{DECAY})$	posterior $P(\text{DS})$	posterior $P(\text{heat-death})$
CPL	25.8%	0.21%	99.79%	0	99.8%
JBP	40.0%	0.79%	99.21%	0	99.2%
BA	58.4%	1.34%	98.66%	0	98.66%
BIN4	50.6%	49.96%	50.04%	0	50.04%

and BA, whereas BIN4 remains close to its native prior fate balance. The residual posterior is therefore strongly grammar-dependent. The divergent-future CPL and JBP families deliver near-certain DECAY; bounded-future BA retains a small RIP component; and frozen-future BIN4 returns 49.96% RIP and 50.04% DECAY, with strict DS mass zero. Its fate-controlling bin is measured as $w_1 = -1.000 \pm 0.044$. Under A-005, 17.6% of BIN4 posterior mass is reported as boundary-adjacent without changing its exact fate label. The binned reconstruction also relocates the finite-redshift signal: the low- z bin is Λ CDM-consistent and the deviation lies at $z \in [0.7, 1.5]$ ($w_3 = -1.53 \pm 0.20$). Thus CPL’s $w_0 \approx -0.88$ is partly the transport of a mid- z signal to $z = 0$ by a linear grammar.

A post-hoc matched-prior audit stops before importance weighting. On the declared seven-point $w(a)$ grid, the CPL, JBP, and BA pushforwards occupy different two-dimensional supports and BIN4 occupies a three-dimensional support: the grid reaches only $z = 1$, so its summary is $(w_3, w_2, w_1, w_1, w_1, w_1, w_1)$ and contains no w_4 coordinate. Their common intersection contains only constant histories and has probability zero under every continuous native prior. A nondegenerate common target is therefore not dominated by all four pushforwards; diagonal covariance regularization would manufacture overlap rather than identify a matched measure. A-004 accordingly terminates the reweighting audit and withholds all matched fate probabilities.

Among the four fitted parametrizations, $P(\text{RIP})$ spans 0.21–49.96% and heat-death compatibility spans 50.04–99.79%. These ranges describe the response of the complete native-prior analyses; they are not a posterior over parametrizations or a universal model-uncertainty interval.

6 Partial identification of the future

6.1 Transported information and the direct-support boundary

Where do the data stop speaking? In the binned grammar the posterior-to-prior information $D_{\text{KL}}[p(w(a))||\pi]$ is 3.03, 2.31, 1.55, and 0.24 nat in the four bins (Fig. 6)—the frozen 0.1-nat horizon is never crossed inside the observed window: the compressed CMB lever arm constrains the highest-redshift bin weakly but above threshold. The registered metric must then be applied to the registered future rule. BIN4 freezes $w(a > 1) = w_1$, so both its posterior and prior are inherited from w_1 and the future KL remains 3.03 nat. Thus a_h is *not reached*; reporting $a_h = 1$ would be inconsistent with the registered metric.

A separate and scientifically important statement does hold: $a = 1$ is the boundary of *direct observational support*. The nonzero KL plotted for $a > 1$ was learned about w_1 at $a \leq 1$ and transported into the future by the BIN4 freezing rule. CPL transports information differently. Thus future fate is not supported by direct likelihood evaluations at $a > 1$, even though the posterior-to-prior KL of the extrapolated function need not vanish.

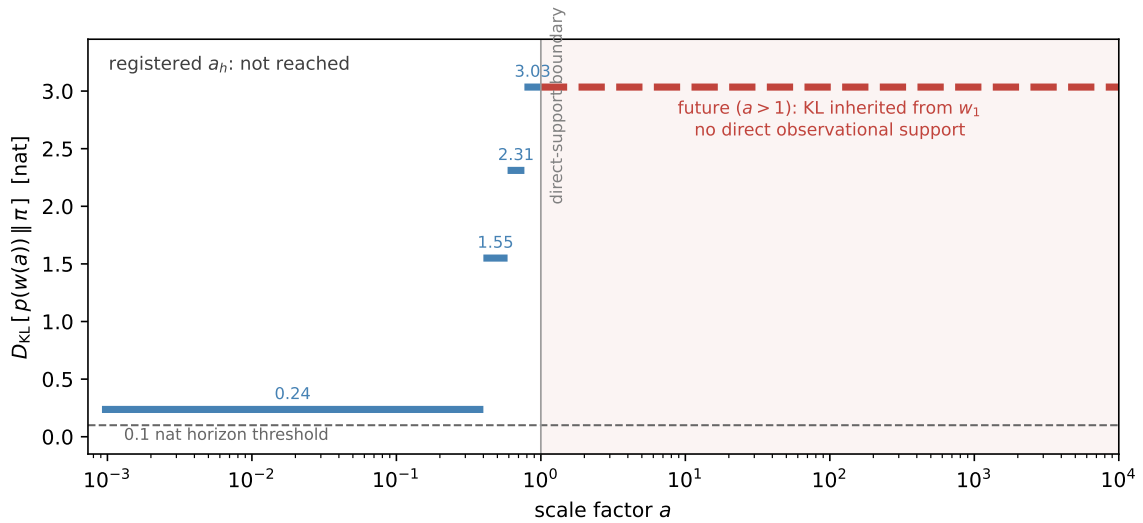


Figure 6. Constraint horizon (BIN4): posterior-to-prior information in $w(a)$ per bin. The dashed future segment is the w_1 KL inherited by the BIN4 freezing rule, not direct future information. The frozen 0.1-nat threshold is never crossed, so the registered result is a_h : not reached. The vertical line at $a = 1$ marks the distinct boundary of direct observational support.

6.2 Direct future-continuation audit

The constraint-horizon calculation diagnoses transported information but does not by itself hold the observed history fixed while changing only the future. The prospectively specified function-space extension performs that stronger test. It fits a common seven-node grammar, with four nodes at $a \leq 1$ and three at $a = 1.5, 2, 4$, using a properly normalized truncated correlated prior. All five registered hyperparameter settings converge, and the inference implementation passes simulation-based calibration on 50 prior- predictive datasets: all datasets converge, all registered 50% and 90% coverage counts pass, and no rank test is rejected after Holm correction.

For the primary setting, the frozen continuation gives $P(\text{RIP}) = 50.63\%$. Across the five settings this moves to 26.76–52.20%. This variation is not evidence that the data measure the remote future node. After conditioning analytically on the four observed-history nodes, the 40-bin posterior-to-prior KL of the final-node residual is only 0.000660 nat in the primary setting and never exceeds 0.000865 nat. The nonzero marginal future-node information is therefore principally a consequence of the global spline and its prior correlations, not an observation at $a > 1$.

The fixed-history continuation audit takes the same 96658 source rows (native weight 308657) and changes only the continuation beyond $a = 1$. The history, every current-background observable, and the likelihood remain identical to machine precision. Table 5 reports the registered families. C0 reproduces the seven-node continuation; C1 holds the present value constant; C2 relaxes smoothly to ΛCDM ; and C3 draws a matched free asymptote. The descriptive envelope is $P(\text{RIP}) = 0.00\% - 50.63\%$. The continuation families have no probability measure, so this range is neither a credible interval nor a model average.

The sharp partial-identification endpoint is stronger than the envelope: 100% of the native source-history weight has support set $\{\text{RIP}, \text{heat}\}$ across the registered continuations.

Thus the observed history alone identifies neither thermodynamic side. This conclusion is conditional on the registered kinematic continuation set; the paths are not asserted to arise from stable, covariant microphysical dark-energy theories.

Table 5. Future-continuation audit. All rows share exactly the same fitted $a \leq 1$ history. C2 is identical for registered transition scales $\tau = 0.5, 1, 2$; the three C3 settings remain approximately balanced.

Family	future rule	$P(\text{RIP})$
C0	frozen FS7	50.63%
C1	present constant	19.72%
C2	relax to $w_\infty = -1$	0%
C3	free asymptote	49.95–50.04%

7 Synthesis and discussion

The reported log-span metrics are collected in Table 7; they are not replaced by the broader eight-axis narrative matrix. The exact zero imposed by P2/P3 makes the prior span infinite. No original-weight F_{data} point is zero: D3 retains a small positive tail even though its equal-weight diagnostic contains zero RIP draws. A finite-run zero is not promoted to a mathematical zero.

Table 6 assembles the audit. For D0 within CPL+P1, the posterior-tail mock rank and the non-equivalent Wilks statistic both lie near the percent level: the 500-mock plus-one value is 0.00599, while Wilks gives 0.024. Across D1–D4, the anti-rip posterior direction remains sub-percent within the same CPL pipeline, but those variants were not separately null-calibrated and the tail magnitude spans 2.665 dex. Quantities specifically about fate—which endpoint, with what probability—also respond strongly to theory-side choices that the current likelihood does not directly constrain.

Three layers. The audit resolves the fate question into three layers with different epistemic status. (i) *Direction*: within the continuous CPL decision problem, the fate-side posterior mass approaches a uniform null distribution when the truth lies on the local Λ CDM boundary. This statement does not transfer automatically to BIN4 or to explicit model averaging. (ii) *Depth*: within CPL+P1 and the adopted compressed-CMB likelihood, the D0 posterior lies in the anti-rip tail of the fitted-truth mock distribution (2/500 null mocks at least as deep, $\hat{p} = 0.00599$, exact 95% interval $[0.000485, 0.0144]$); the non-equivalent Wilks test gives $p = 0.024$. The off-boundary experiment additionally shows strong direction power at the two registered outer truths, but materially weaker inner-point direction and depth power. This is a pipeline-conditional finite-redshift preference, not a calibration of data systematics, likelihood compression, or cross-model uncertainty. Across D1–D4 the anti-rip posterior direction remains sub-percent within CPL+P1, but these variants were not separately null-calibrated. Separately, BIN4 places its largest in-window departure at $z \in [0.7, 1.5]$. (iii) *Translation*: converting finite-redshift information into a fate probability additionally requires an extrapolation rule and induced prior measure that observations do not directly constrain. The exact continuation audit makes this last layer constructive: without altering any current-background observable or likelihood value, every weighted source history admits both rip and heat-death-compatible registered continuations.

Why the fragility is structural, not accidental. The fate map is a *discontinuous* functional of parameters that observation constrains only *continuously*: RIP and DECAY are separated by

Table 6. The eight-axis fragility matrix. Each row varies one audit package or axis and reports the response of the fate verdict. The grammar row necessarily couples functional form, native dimension, and its registered coordinate prior.

Axis	knob	response	verdict
SN compilation	Pantheon+/Dovekie/Union3	Λ CDM exclusion $2.25 - 3.27\sigma$	magnitude fragile
SH0ES anchor	in/out	$H_0 + 1.37 \text{ km s}^{-1} \text{ Mpc}^{-1}$; $w_0 - 0.036$	leakage; direction stable within CPL
CMB information	CPL compressed/full benchmark	stable -0.5σ discount	BIN4 remains compressed-only
Dataset ablation	D1–D4	$w_a \in [-1.72, -0.55]$; $P(\text{RIP}) 0.0008 - 0.376\%$	direction stable; tail 2.665 dex
Prior	P1/P2/P3	$P(\text{RIP})$: $0.18\% \rightarrow 0$ (excluded)	prior decides
Statistic	$\Delta\chi^2$ vs $\ln B$	2.25σ against vs. $\ln B = -1.617$ (CPL/ Λ CDM)	conditioning flips
Grammar	four native-prior fits	$P(\text{RIP}) 0.21 \rightarrow 49.96\%$	dominant axis
Truth position	Λ CDM in continuous CPL	verdict $\sim U(0, 1)$; zero-residual 52/48	conditional null law

Table 7. Reported fragility measures, in dex. The parametrization row covers the four converged native-prior fits.

Axis	$P(\text{RIP})$ span	$P(\text{heat})$ span
F_{prior}	∞	0.002
F_{param} , fitted four	2.380	0.300
F_{data}	2.665	0.002

a zero-width boundary, and Λ CDM sits on it. Under regular continuous-model asymptotics, repeated-data class probabilities remain uniform when the truth is exactly on a locally linear boundary, even as the likelihood narrows. This statement is conditional: a discrete Λ CDM component, model averaging, nonregular priors, or a physically derived theory changes the inferential object. In that qualified sense the result is a quantitative instance of the underdetermination of dark energy articulated in Refs. [29, 30], not a no-go theorem for every future Bayesian procedure.

What would change the situation. More precise distance data can narrow the finite-redshift function and may reveal that the truth is off the boundary, but observations made today cannot directly sample $a > 1$. The registered KL horizon is not reached; the relevant limitation is instead the direct-support boundary at $a = 1$. Extrapolation is licensed by trusted theory, not by fit quality alone. CPL and its cousins are more than raw curve fitting—they are compressions calibrated against quintessence-class dynamics inside the window [31–33]—but a calibrated compression is still not a theory: the calibration certifies fidelity to a model class over the fitted range, and extending that fidelity to $a > 1$ is precisely the adoption of the class as a prior—an axis this audit prices (Sec. 5.2). A physically derived dark-energy model (a specific potential, a symmetry argument) would convert fate back into a derivation—then data would test the theory inside the window and the theory would carry the license beyond it. Until then, fate probabilities remain conditional predictions of the adopted model class.

The continuation result should be read at the same level. C0–C3 are smooth background expansion rules, not demonstrations that every path arises from a stable covariant field theory with acceptable perturbations. A specified microphysical theory may link the present state to a unique future through its equations of motion and thereby exclude most or all of these continuations. Our result therefore establishes partial identification when no such dynamical license is supplied; it does not prove that cosmic fate is unknowable in every physical theory.

Relation to the recent fate literature. Papers reporting $P(\text{fate})$ have always been conditional on a parametrization or physical model. Earlier work explicitly replaced CPL with a bounded future parametrization, and recent studies examine multiple parametrizations

or specific scalar-field scenarios [10–12, 28]. These are scientifically meaningful conditional calculations. Our narrower contribution is to report the induced prior fate mass, cross-grammar sensitivity, and fitted-truth null calibration in one audit. Table 6 is therefore a finite, non-exhaustive sensitivity budget, not a universal model-uncertainty interval. The framework—local preregistration, null calibration, and declared assumption axes—can be adapted to other inferences whose conclusions lie beyond their data window.

Limitations. All significances inherit the compressed-CMB condition (priced at $\approx 0.5\sigma$; A-001). That condition is mildest for CPL-like histories, for which the distance priors were validated [21]; for the binned grammar, whose high- z bin can depart further from Λ , extraction-side validity is an additional inherited assumption. The analysis is background-only: it does not evolve CMB spectra, dark-energy perturbations, or late-time growth. Growth information could shift the finite-redshift posterior, but it would not directly observe $a > 1$. For BIN4, we impose the hard gate $\rho_{\text{DE}}/\rho_m(z = 1059) < 0.01$; the archived posterior has zero weighted violating mass and a maximum sampled ratio 8.62×10^{-5} ; this check does not turn the distance prior into a full-CMB likelihood.

The four converged parametrizations are neither exhaustive nor a model average. Their native priors induce different measures on functions and fates, and their pushforwards have incompatible supports on the declared common summary. Consequently, their probability span is a sensitivity diagnostic, not a credible interval or universal model-uncertainty bound. The repeated- data calibration is likewise conditional on fitted-truth Λ CDM analyzed with CPL+P1; it does not calibrate model misspecification or every grammar. The registered continuations are kinematic, and heat-death compatibility is a coarse endpoint grouping rather than a calculation of vacuum decay, black-hole evaporation, or quantum-gravitational thermodynamics. Further parametrizations and physical reconstructions remain possible [28]. The work was performed by a single author with AI assistance under the verification protocol described; residual unknown-unknowns are bounded only by the public audit trail.

8 Conclusions

Within CPL+P1 and the adopted compressed-CMB likelihood, the data give a pipeline-conditional finite-redshift result: 2 of 500 fitted-truth Λ CDM mocks are at least as deep as the real CPL tail, giving $\hat{p} = 0.00599$, while the distinct Wilks test gives $p = 0.024$. This calibration is informative about the specified CPL decision problem, not about every model or data systematic.

Translation from that result to cosmic fate is assumption-dependent. A quintessence prior excludes the rip before data are considered, and four converged parametrizations, each with its native prior, move posterior $P(\text{RIP})$ from 0.21% to 49.96% against already different induced prior masses. These values diagnose sensitivity; they are not a common-prior model comparison.

The direct-support boundary is $a = 1$. In the calibrated seven-node analysis, the final-node conditional-residual information is only 0.000660 nat. Holding the entire observed history fixed, the registered continuations span $P(\text{RIP}) = 0.00\% - 50.63\%$, and 100% of source-history weight admits both fate sides. No probability is assigned to the continuation families. Thus present observations can produce sharp fate probabilities within a chosen model, but they do not identify a unique infinite future without a physical continuation law. This is a background-level partial- identification result within the registered kinematic class, not an impossibility theorem for all microphysical theories.

Acknowledgments

This work used CAMB [34], cobaya [35], GetDist [36], and dynesty [37]; we thank their authors, and the DESI collaboration for the public release of the DR2 cosmology chains used in validation. Generative-AI assistance was provided by Anthropic Claude Fable 5 and OpenAI Codex (GPT-5) for portions of pipeline implementation, literature triage, manuscript drafting and editing, and reproducibility and consistency checks. All AI-assisted material was critically reviewed under the version-controlled verification protocol described in Sec. 2; the tools did not generate or alter original research data, and all analysis choices, amendment sign-offs, and final claims are the author’s responsibility.

A Analysis plan and reported scope

The locally frozen original under tag `prereg-v1` remains archived in the repository and takes precedence over this summary. It was not deposited in a third-party registry at freeze time. This appendix records only the components that contribute to the reported scientific results; amendments are summarized in Appendix B.

Research question. The declared deliverable was a set of fragility measures, not a fate discovery. Data-dominated and assumption-dominated outcomes were both designated as publishable before the first target fate calculation.

Model and data. The analysis treats background expansion only, $H^2(a)/H_0^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_{DE} f_{DE}(a)$, with flat CPL as the baseline and BA, JBP, and BIN4 as the converged parametrization-sensitivity set [38, 39]. D0 combines DESI DR2 BAO, the CMB distance prior, and Pantheon+ with SH0ES; D1–D4 remove one component at a time, and the supernova-compilation audit replaces Pantheon+ while holding the other choices fixed. A prospectively specified extension adds the calibrated seven-node reconstruction and fixed-history continuation audit.

Priors and fate rules. P1 is the baseline native-coordinate prior; P2 adds pointwise $w(a) \geq -1$; P3 adds the phenomenological thawing restrictions. The classifier priority is CRUNCH > RIP > DS > DECAY > OTHER, with a 1% OTHER budget and a doubled/halved-threshold boundary audit. The operational thermodynamic map groups DS and DECAY as heat-death-compatible and RIP and CRUNCH as non-heat-death.

Statistical and reporting rules. Chains entering results must satisfy $R - 1 < 0.01$ on the worst parameter. Sub-percent fate tails require independent nested-sampling verification. The reported sensitivity measures are the $\log_{10}$ spans across P1–P3, the four converged parametrizations, and D0–D4. Repeated-data calibration, model evidence, the constraint-horizon diagnostic, and exact continuation invariance are reported as distinct inferential objects rather than interchangeable significances.

Freeze discipline. The tagged plan is immutable. Changes are recorded in an append-only ledger with their trigger and timing relative to data contact. Exploratory extensions that do not satisfy their completion gates are excluded from scientific results; their audit records remain in the repository.

B Amendment log

The complete signed ledger is archived in the repository. The entries below summarize only changes that affect the interpretation or reproducibility of reported results.

A-001 (2026-07-03; before fate analysis). Gate 1 was changed from matching the official full-*Planck* exclusion significance to matching published compressed-CMB analyses of the same class. The best-fit-point gate was unchanged. The pipeline agrees with the official contour, but every reported significance inherits the measured $\approx 0.5\sigma$ compression discount (Secs. 2.1 and 5).

A-002 (2026-07-06; during the mock campaign). The invalid requirement that at least 90% of boundary-truth mocks favor heat-death compatibility was replaced by reporting the null distribution itself. For regular continuous CPL at the Λ CDM boundary, the predicted class probability is approximately $U(0, 1)$; the completed campaign confirms that conditional result (Sec. 4). The OTHER and boundary budgets were unchanged.

A-003/A-005 (2026-07-06/21; classifier correction). The registered OTHER budget exposed a gap for parametrizations with finite w_∞ . An analytic finite-limit branch was added and later corrected to the physical boundaries: RIP iff $w_\infty < -1$, DS iff $w_\infty = -1$, and DECAY iff $w_\infty > -1$; $\epsilon = 0.01$ is only a boundary-adjacent flag. BA and BIN4 were reclassified accordingly. CPL, JBP, D0–D4, P2/P3, the mock calibration, Wilks statistics, and evidence are unchanged.

A-004 (2026-07-21; post-hoc support audit). On the declared seven-node summary, CPL, JBP, and BA occupy different two-dimensional supports and BIN4 occupies a three-dimensional support. Their common intersection contains only constant histories and has probability zero under every continuous native prior. No nondegenerate common reweighting target therefore exists; matched fate probabilities are withheld (Sec. 5.4).

A-006/A-007 (2026-07-21/22; scope and provenance). The insufficient condition $w_4 < 0$ was replaced by $\rho_{\text{DE}}/\rho_m(z = 1059) < 0.01$. The archived BIN4 posterior has zero weighted violating mass and a maximum ratio 8.62×10^{-5} . The ledger also records the grammar-specific translation of the original early-matter-domination intent. No reported probability changes under these records, and BIN4 remains background-only and compressed-CMB-conditioned.

Analyses that did not pass their completion gates are absent from the scientific results; their exclusion records remain in the repository.

C Exploratory in-window fit audit

Table 8 reports the lowest likelihood χ^2 values in the archived post-burn-in chains. Common parameters cancel in ΔAIC . The entries are descriptive chain minima, not dedicated optimizations, evidences, or model probabilities.

Table 8. Exploratory in-window fit audit.				
Grammar	$\chi^2_{\text{chain,min}}$	$\Delta\chi^2$	ΔAIC	$R - 1$
CPL	1489.534	0.000	0.000	0.0063
JBP	1492.075	2.541	2.541	0.0093
BA	1491.523	1.989	1.989	0.0026
BIN4	1485.386	−4.149	−0.149	0.0046

D Classifier numerics and known edges

The classifier integrates each posterior sample’s background from $a = 1$ to $a_{\text{max}} = 10^4$ and applies Sec. 2.2’s criteria with the frozen thresholds: RIP tail-convergence tolerance 10^{-3} (relative change of $t(a)$ from a_{max} to $2a_{\text{max}}$); DS tolerance $|w(a_{\text{max}}) + 1| < 0.01$; DECAY thresholds

$\rho_{\text{DE}}(a_{\text{max}})/\rho_{\text{DE}}(1) < 0.01$ or $\rho_{\text{DE}}/\rho_m(a_{\text{max}}) < 1$; A-005 finite-limit boundary-adjacent tolerance $\epsilon = 0.01$. Numerics: the age integral $\int da/(aH)$ is evaluated by adaptive quadrature on 40 logarithmically spaced panels; the CRUNCH scan uses a 400-point logarithmic grid in a ; $\ln f_{\text{DE}}$ is capped at 700 before exponentiation to avoid overflow (samples beyond the cap are already deep in the RIP regime). The boundary flag re-runs the classification with all thresholds doubled and halved and marks any sample whose label flips.

The implementation passes a ten-case unit suite: $\Lambda\text{CDM} \rightarrow \text{DS}$ (unflagged); phantom $w_a > 0 \rightarrow \text{RIP}$; the DR2 best fit $\rightarrow \text{DECAY}$; mild thawing $\rightarrow \text{DECAY}$; a closed, decaying-DE case $\rightarrow \text{CRUNCH}$; finite-limit cases just below, above, and exactly at $w_\infty = -1$ map to RIP, DECAY, and DS, respectively, with the boundary-adjacent flag raised; and two *documented limitation cases* of the frozen criteria, constant- w curves with $w = -0.98$ (quasi- Λ : neither DS-tight nor decayed at a_{max}) and $w = -1.2$ (power-law phantom: a true finite-time singularity whose tail convergence at $a_{\text{max}} = 10^4$ is slower than the frozen tolerance), both of which are intentionally routed to OTHER, i.e. to the manual-audit channel. Constant- w curves are measure-zero in the CPL posterior; the posterior mass in the band $|w_a| < 3 \times 10^{-4}$ is $\sim 10^{-3}$, safely inside the 1% OTHER budget. These edges were documented before the real-data runs and did not require an amendment; the distinct gap later exposed by bounded-future grammars did (A-003, Appendix B).

Across the four reported parametrizations, the cross-grammar machinery takes r_{drag} from a ΛCDM computation (drag-epoch physics treated as dark-energy independent); this is self-consistent a posteriori even for the binned grammar, whose free high- z bin sits at $w_4 = -1.71 \pm 0.62$. The A-006 weighted audit gives zero posterior mass above $\rho_{\text{DE}}/\rho_m(z = 1059) = 0.01$ and a maximum sampled ratio 8.62×10^{-5} ; this posterior check does not turn the compressed distance prior into a full-Planck likelihood.

E Mock machinery and zero-residual validation

The null-calibration truth is the ΛCDM best fit to D0 ($\Omega_m = 0.2968$, $H_0 = 68.81$, $M_B = -19.397$), located by repeated bounded minimization. Each noisy mock draws one noise realization per likelihood from the exact covariances used in inference—SN 1657×1657 (stat+sys, retaining rows with $z_{\text{HD}} > 0.01$ or a Cepheid calibrator), BAO 13×13 , CMB prior 3×3 —added to the truth’s predicted data vectors, and is then processed by the full CPL+P1 pipeline (MCMC and classifier identical to the real-data runs). The campaign is a point null—mocks are generated at the single best-fit truth rather than marginalized over the ΛCDM posterior. The historical v1.x checkpoint contained noisy indices 1–100. The frozen extension replayed the same generator stream and appended indices 101–500 in an isolated result tree; it could not stop early and did not overwrite the v1.x inputs. For the direction layer this is sufficient: the uniformity prediction is truth-value-independent for any truth on the fate boundary (Appendix F), and every ΛCDM parameter point sits on it; the depth-layer floor inherits the usual parametric-bootstrap caveat of being conditioned on the fitted truth. Mocks are generated deterministically from a single seed, so the entire campaign regenerates bit-identically from the repository. A 101st, zero-residual realization (mock000) provides a closed-loop test: at the truth point all three likelihood χ^2 vanish identically, verifying the mock-likelihood round trip; its fate verdict (52/48) is the finite-covariance boundary diagnostic of Sec. 4.

Six falsifiable predictions were logged during the original campaign (after ~ 17 of its 100 mocks had been inspected), derived from the probability-integral-transform argument, before

the campaign’s completion and before the zero-residual realization ran: direction accuracy $50 \pm 10\%$; P_{heat} consistent with $U(0,1)$; 0–3 of 100 mocks below the real data’s $P(\text{RIP})$; OTHER and boundary budgets at the 10^{-3} level; `mock000` near 0.5; and the resulting A-002 trigger. All six were correct at that historical checkpoint. The extension explicitly declined to predict that 0/100 would remain zero: its completed result is 2/500. Per-mock records (501 rows), the completion audit, the six versioned endpoint blocks, and the batch logs are in the repository.

F Boundary calibration statistics

This appendix collects, in an idealized regular-model form, the statistical facts behind the null calibration of Sec. 4: the exact uniformity result, its closed-form verification, the diagnostics that detect a *miscalibrated* pipeline, the finite-mock reporting rules, and the relation to the calibration literature.

One dimension. Let $\hat{x} \sim \mathcal{N}(\theta_0, \sigma^2)$ with posterior $\theta | \hat{x} \sim \mathcal{N}(\hat{x}, \sigma^2)$ (the calibrated case), and let class A be $\theta > 0$ with the truth on the boundary, $\theta_0 = 0$. Then $P(A | \hat{x}) = \Phi(\hat{x}/\sigma) = \Phi(Z)$ with $Z \sim \mathcal{N}(0,1)$: by the probability integral transform [40], the class probability is *exactly* $U(0,1)$ across realizations. Numerically (2×10^5 replications): mean 0.5010, variance 0.08356 against the uniform value $1/12 \approx 0.08333$, direction rate 0.502, KS $p = 0.16$. Because the class probability is a continuous functional of the data, its null distribution has no atom at $1/2$: individual realizations can look arbitrarily confident while the ensemble verdict remains a fair coin.

Two dimensions, correlated errors. In a (u,v) analogue of (w_0+1, w_a) with a D0-like degenerate covariance and a linear fate score $s = \alpha u + v$ ($\alpha = 1.2$, class $s > 0$), the same argument applies to any linear boundary through the truth: 5×10^4 replications give mean 0.5013, variance 0.08326, direction rate 0.505, KS $p = 0.058$ (at 5×10^4 replications the KS test resolves sub-percent distortions; this is comfortable agreement). The CPL fate boundary $w_a = 0$ is exactly linear, so the result covers the geometry of Sec. 4 up to the non-Gaussianity of the real posterior—which the full-pipeline campaign tests directly. The derivation assumes a continuous posterior with no atom on the boundary. It does not apply unchanged to spike-and-slab priors or Bayesian model averaging that assigns positive probability to exact ΛCDM .

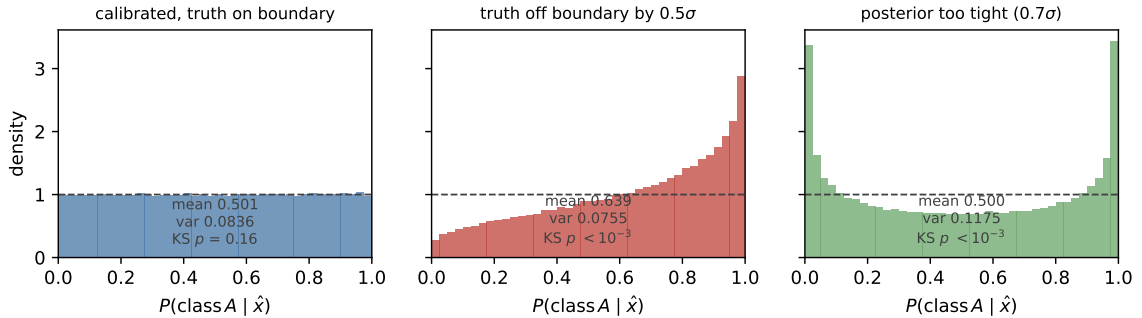


Figure 7. The boundary-calibration toy (Appendix F). Left: calibrated posterior with the truth on the boundary—class probabilities exactly uniform. Center: truth off the boundary by 0.5σ —direction information appears, as it should. Right: over-confident posterior—mean preserved but variance above $1/12$; the uniformity diagnostics fail precisely when the pipeline is defective.

Diagnostics: when uniformity must fail. The value of the null test lies in what breaks it (Fig. 7). Truth off the boundary by 0.5σ : mean 0.639, direction rate 0.692, KS $p \approx 0$ —direction information appears exactly when it exists. Decision threshold shifted while the truth sits on it: still uniform (KS $p = 0.25$)—only the truth–boundary distance matters. Posterior too wide (under-confident, 1.5σ): mean stays at 0.500 but the variance collapses to $0.0496 < 1/12$. Posterior too tight (over-confident, 0.7σ): variance inflates to $0.1175 > 1/12$. Strongly curved boundary: KS $p = 2.5 \times 10^{-6}$ —uniformity is a locally-linear statement (the CPL boundary is exactly linear). The triple observed in Sec. 4—accuracy $\approx 50\%$, uniform shape, variance consistent with $1/12$ —shows no gross defect of the tested kinds. The diagnostics respond in distinct directions in the simulations, but a 100-realization campaign retains limited power against mild misspecification.

Finite-mock reporting rules. With k of n fitted-truth mocks at or below the real-data statistic, we report the conditional plus-one estimate $\hat{p} = (k+1)/(n+1)$ [41] and the exact two-sided 95% Clopper–Pearson interval for the underlying tail probability; a finite-simulation p -value is never reported as zero. At the historical $k = 0, n = 100$ checkpoint these rules gave $\hat{p} = 0.0099$ and upper bound 0.0295. The completed extension has $k = 2, n = 500$, $\hat{p} = 0.00599$, and interval $[0.000485, 0.0144]$. The larger campaign strengthens finite tail resolution while invalidating the historical “below every mock” wording.

Relation to the calibration literature. The lineage runs from the probability integral transform [40] through prequential calibration [42] and posterior predictive checking [43, 44]. The closest modern relative is simulation-based calibration [45], which validates posterior computation via rank uniformity under prior-drawn truths. The present test differs in both conditioning and target: the truth is *fixed at the boundary point*, and the calibrated object is a *decision functional discontinuous there*—the campaign is a null test of the verdict, not of the posterior machinery (which Gate 1 validates separately). The practical residue is a reporting discipline matching the three layers of Sec. 7: a fate-style claim should state (i) its direction-layer verdict together with the null distribution of that verdict, (ii) its depth-layer tail probability under the finite-simulation rules above, and (iii) the translation-layer conditions—parametrization, prior, statistic—under which the verdict holds; Table 6 is the translation layer’s error budget.

G Data, code, and reproducibility

The repository, https://github.com/shlinawaf717-collab/cosmo_fate (archived at Zenodo, <https://doi.org/10.5281/zenodo.21332433>), contains the frozen plan and amendment log, the full pipeline (likelihood configurations, fate classifier and its unit suite, mock generator, batch drivers, nested verification), archived thin chains and per-mock records sufficient for the reported summaries, the figure scripts, and deterministic audits for induced prior fate mass, in-window chain minima, model averaging, the registered constraint-horizon metric, the matched-prior support audit, and the grammar-prior implementation trace. The extension additionally versions the 500-mock completion and endpoints; the six truth-specific power fits, completion audit, exact power table, and profile- χ^2 -annotated figure; the seven-node SBC and five-setting endpoints; and the exact fixed-history continuation audit. All input data are public: DESI DR2 BAO compressed measurements and covariance; Pantheon+ (with SH0ES), Union3, and DES-Dovekie supernova files obtained via the `cobaya` installer; the CMB distance prior of Ref. [21] (Table I, Λ CDM row, *Planck* 2018 TT,TE,EE+lowE), entered by value; and the official DESI DR2 w_0w_a CDM chains used in Gate 1, downloaded

from the public DESI data portal. Every input file is registered in the repository manifest with source, version, and SHA-256 checksum.

Software: Python 3.13, `cobaya` 3.6.2, `camb` 1.6.6, `dynesty` 3.0.0, `getdist` 1.7.7, `numpy` 2.5.0, `scipy` pinned to 1.16.2 (a `scipy` 1.18 regression in spline evaluation breaks `camb` 1.6.6’s BBN interface; the pin and the upstream fix are documented in the manifest). The classifier unit suite, the zero-residual closed-loop test, deterministic mock regeneration, and CI audit invariants together allow every number in this paper to be reproduced from the repository; the environment setup, data installation, and per-result commands are documented in the repository README.

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